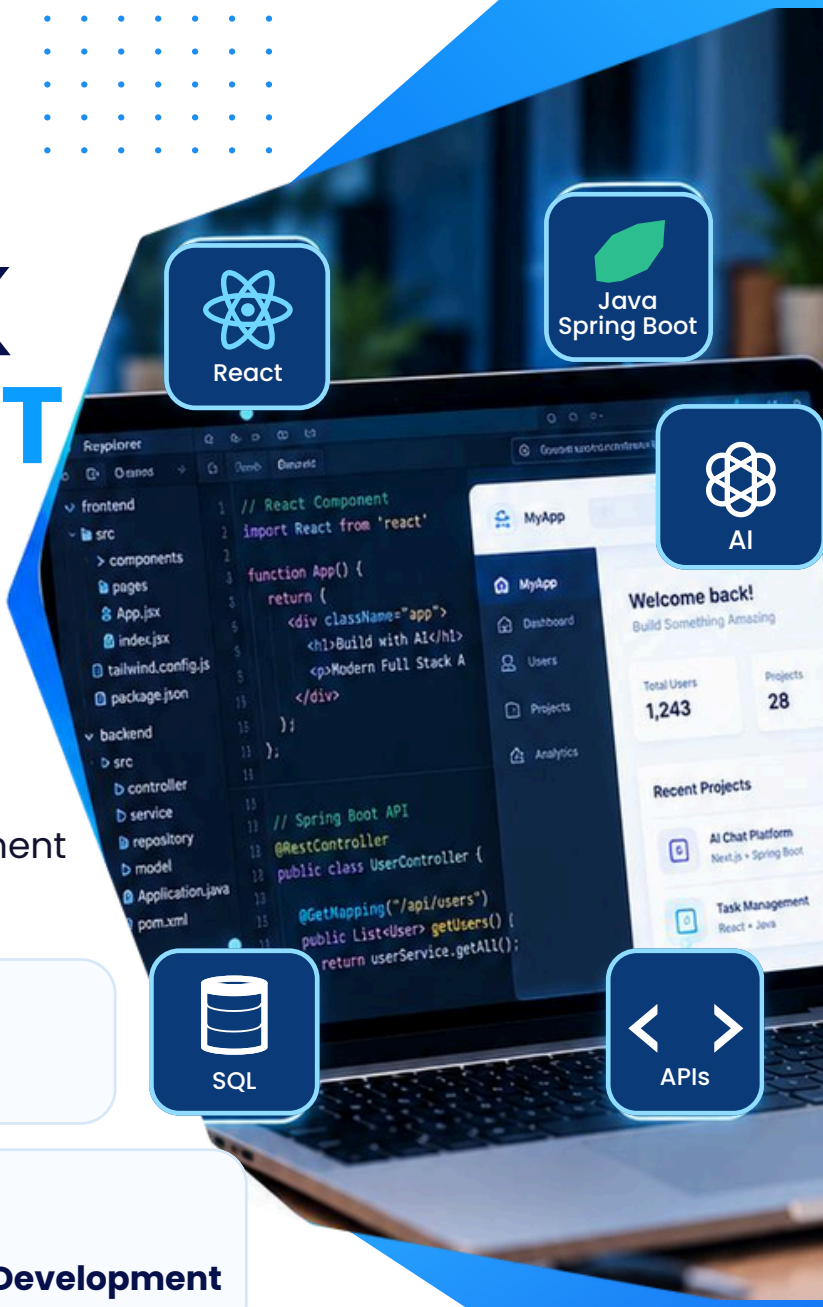
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CREATIVE LABS
Learn. Create. Build. Beyond.

FULL STACK DEVELOPMENT WITH AI



3-MONTH INDUSTRY-SUPPORTED CAREER PROGRAM

Learn to build modern, full-stack web applications with AI-powered development workflows.

DURATION
3 Months / 12 Weeks

LEARNING MODE
Technical Training + Hands-on Learning + Project Development

LEARN MODERN TECH. BUILD REAL PROJECTS. CREATE AI-POWERED APPLICATIONS.

2 MINI PROJECTS + 1 CAPSTONE PROJECT

INDUSTRY-RELEVANT TOOLS & TECHNOLOGIES

CAREER-READY SKILLS

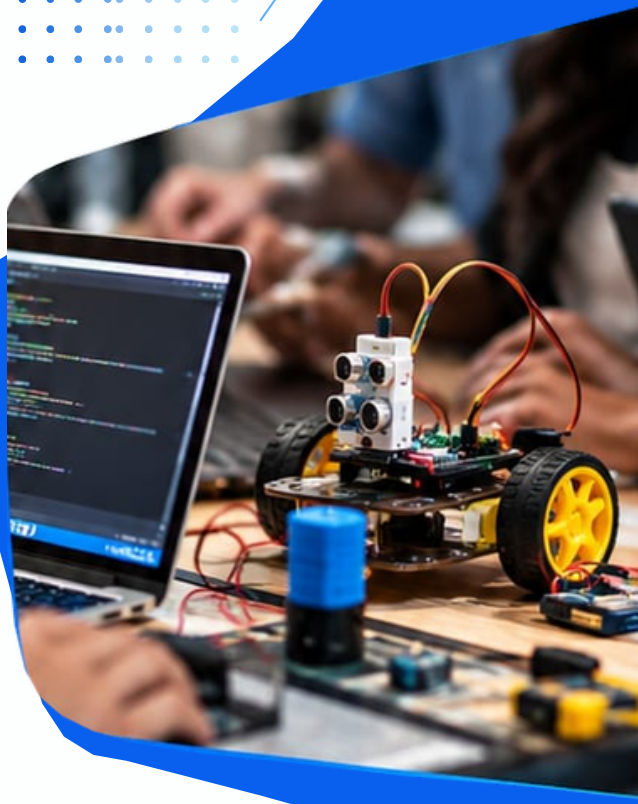
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TECHKNOTS
One knot at a time

A B O U T

IMPARA

POWERED BY TECHKNOTS



Learn. Create. Build. Beyond.

Impara Creative Labs is a student-focused learning and career development platform that helps students turn technical knowledge into practical skills. Through hands-on training, project-based learning, mentorship and industry exposure, Impara helps students learn, build and prepare for real-world technology careers.

POWERED BY TECHKNOTS
Built on the technology, R&D and industry experience of TechKnots.

W H A T W E D O

Industry-Oriented Learning
Practical learning built around relevant skills, tools and technologies used in real-world development.

Hands-On Projects
Learn by building practical projects and applying concepts.

Industry Exposure
Understand real-world workflows, tools and project environments.

Career Preparation
Build technical skills, project experience and professional readiness.

O U R F O C U S A R E A S

- IoT & Smart Automation
- Full Stack Development
- AI & Generative AI
- Data & Emerging Technologies

P O W E R E D B Y T E C H K N O T S

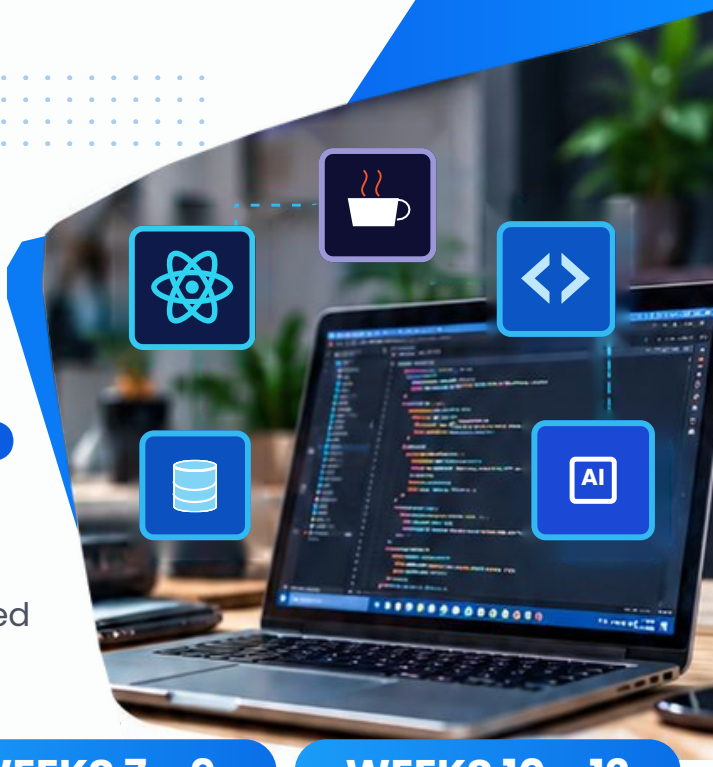
- R&D & Technology
- Industry Experience
- Technology Innovation
- Student Ecosystem

Impara helps students build real-world skills, not just learn theory.

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12-WEEK
COURSE ROADMAP

A structured learning journey designed to take you from web fundamentals to real-world, project-based full-stack application development.



WEEKS 1 - 3

WEB & PROGRAMMING FOUNDATIONS

- HTML & CSS
- Responsive Web Design
- JavaScript
- DOM & Events
- JSON & Fetch
- Async JavaScript
- Core Java
- OOP Fundamentals

WEEKS 4 - 6

FRONTEND & DATABASE

- Java Collections
- React
- Components & JSX
- Props & State
- Forms & Routing
- Tailwind CSS
- SQL
- Database Design
- CRUD & Joins
- **Mini Project**

WEEKS 7 - 9

BACKEND & APPLICATION DEVELOPMENT

- JDBC
- Database Connectivity
- Spring Boot
- Controllers & Services
- Repositories & DTOs
- REST Endpoint Design
- Hibernate
- ORM & Persistence
- Database Relationships

WEEKS 10 - 12

INTEGRATION, AI & FINAL APPLICATION

- REST API Integration
- React → Spring Boot → SQL
- Authentication & Authorization
- Testing & Validation
- AI Integration
- Error Handling & Security
- Production Readiness
- Documentation
- Final Demonstration

LEARNING JOURNEY

LEARN → **BUILD** → **INTEGRATE** → **DEPLOY**

LEARN
Build strong foundations in web development, Java and programming.

BUILD
Develop frontend, database and backend components.

INTEGRATE
Connect React, REST APIs, Spring Boot and SQL.

DEPLOY
Test, document and demonstrate a complete full-stack application.

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TECHNOLOGIES & TOOLS



Work with industry-relevant technologies and tools to build real-world full-stack applications with AI.

01 FRONTEND DEVELOPMENT

- HTML
- CSS
- JavaScript
- Tailwind CSS
- React

02 BACKEND DEVELOPMENT

- Core Java
- Object-Oriented Programming
- Collections
- JDBC
- Hibernate
- Spring Boot

03 DATABASE

- SQL
- Relational Databases
- Database Design
- CRUD Operations
- Joins
- Transactions

04 API & INTEGRATION

- REST APIs
- HTTP
- JSON
- Frontend - Backend Communication

05 AI INTEGRATION

- OpenAI API
- LangChain
- Generative AI
- AI-Assisted Application Development

06 DEVELOPMENT ENVIRONMENT

- VS Code
- GitHub
- Docker
- Postman

07 ADDITIONAL TOOLS & FRAMEWORKS

- Node.js
- MongoDB
- Maven / Gradle
- Other Development Tools

USE THESE TOOLS TO

LEARN
Work with industry-standard tools and technologies.

BUILD
Develop real-world frontend, backend and database components.

INTEGRATE
Connect APIs, databases and AI into a full-stack application.

DEPLOY
Test, optimise and demonstrate production-ready applications.

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SKILLS YOU'LL BUILD

Gain in-demand skills to design, develop and deploy real-world full-stack applications with AI.



01 FRONTEND DEVELOPMENT

- Build modern, responsive web interfaces
- Work with HTML, CSS, JavaScript, Tailwind CSS
- Develop dynamic UIs with React
- Implement forms, validation and routing

02 BACKEND DEVELOPMENT

- Learn Core Java and OOP concepts
- Work with Collections, JDBC and Hibernate
- Build REST APIs with Spring Boot
- Implement business logic, validation and exception handling

03 DATABASE DEVELOPMENT

- Design and work with relational databases
- Write SQL queries and perform CRUD operations
- Work with joins, constraints and database design
- Connect applications using JDBC and Hibernate

04 API & APPLICATION INTEGRATION

API

- Build and consume REST APIs
- Work with HTTP, JSON and API integration
- Connect frontend and backend applications
- Understand real-world application architecture

05 AI INTEGRATION

- Integrate AI capabilities into full-stack applications
- Work with OpenAI API and LangChain
- Build AI-assisted features and workflows
- Explore real-world use cases of Generative AI

06 DEVELOPMENT PRACTICES

- Use professional development tools and workflows
- Work with Git and GitHub
- Test and debug applications
- Follow best practices for clean, maintainable code
- Prepare for real-world deployment

CAREER ROLES

Be prepared for in-demand roles in software development and AI.

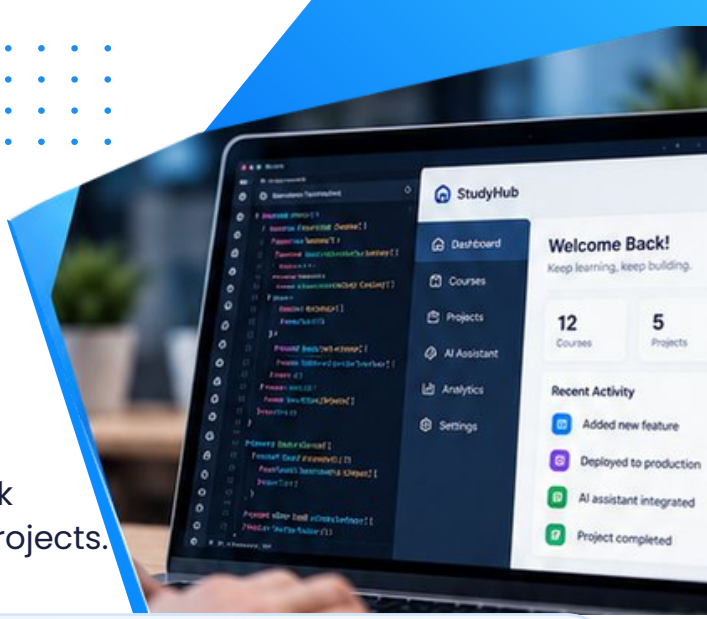
- Full Stack Developer
- Frontend Developer
- Backend Developer
- Java Developer
- React Developer
- Spring Boot Developer
- Web Application Developer

Build the skills. Explore the opportunities.
Be ready for real-world technology careers.

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PROJECT-BASED LEARNING

Apply what you learn to build real-world, full-stack applications with AI features through hands-on projects.

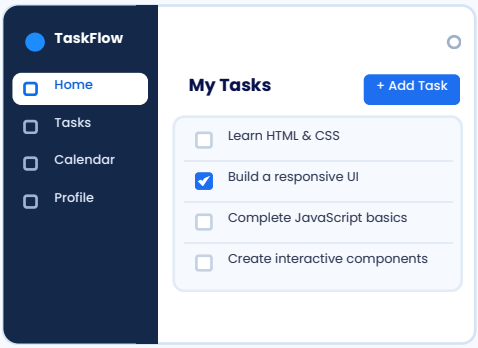


2 MINI PROJECTS + 1 CAPSTONE PROJECT

A progressive learning experience to help you apply frontend, backend, database, API and AI concepts, and create a complete full-stack application.

01 MINI PROJECT 1

WEEKS 1 – 4

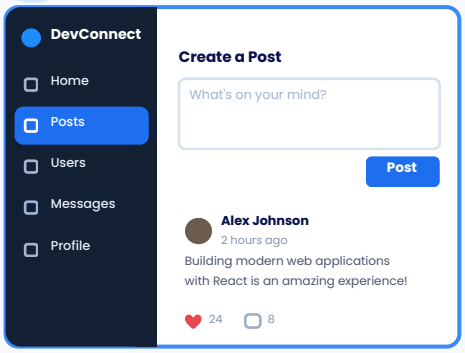


- Build a simple web application using HTML, CSS and JavaScript
- Implement form handling and validation
- Work with DOM, events and dynamic content
- Deploy the application with a user interface

TECHNOLOGIES

02 MINI PROJECT 2

WEEKS 5 – 8

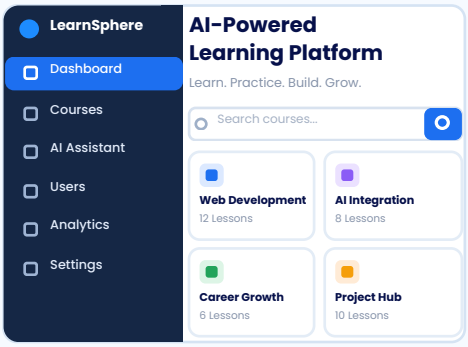


- Build a React-based application
- Create reusable components with Tailwind CSS
- Implement routing and form handling
- Integrate with a backend API (using mock API or basic backend)

TECHNOLOGIES

03 CAPSTONE PROJECT

WEEKS 9 – 12



- Build a complete full-stack application with React and Spring Boot
- Implement database integration (SQL)
- Develop and consume REST APIs
- Integrate AI features (Intel GenAI)
- Deploy and present a real-world solution

TECHNOLOGIES

WHAT YOU'LL PRACTICE

- Frontend Development
- Backend Development
- Database Design & Integration
- API Development & Integration
- AI Feature Integration
- Deployment & DevOps

3 HANDS-ON PROJECTS

1 COMPLETE FULL-STACK APPLICATION

From web fundamentals to a real-world, AI-powered full-stack application.



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BEYOND THE CLASSROOM

Gain industry exposure, build a strong portfolio and get career-ready with our comprehensive support system.



01

PORTFOLIO-READY PROJECTS

Build 2 mini projects + 1 capstone project to showcase your skills.

02

INDUSTRY EXPOSURE

Learn from industry experts through live sessions, real-world use cases and the latest tools and trends.

03

EXPERT MENTORSHIP

Get guidance from industry professionals throughout your learning journey.

04

CERTIFICATION

Receive a course completion certificate to validate your skills and project experience.

05

LINKEDIN PROFILE SUPPORT

Get help in optimising your LinkedIn profile to showcase your skills, projects and achievements effectively.

06

MOCK INTERVIEWS

Practice with mock interviews to build confidence and prepare for real opportunities.

CAREER OPPORTUNITIES

This program prepares you for exciting opportunities in full-stack development and related domains.

- Full Stack Developer
- Frontend Developer
- Backend Developer
- Java Developer
- React Developer
- Spring Boot Developer
- Web Application Developer
- Software Engineer
- AI-Integrated Developer
- Technical Consultant

Explore opportunities across startups, product companies and research organisations.

YOUR JOURNEY WITH IMPARA

LEARN
Industry-relevant skills with hands-on training

BUILD
Work on real-world full-stack projects

SHOWCASE
Create a strong portfolio and LinkedIn profile

PREPARE
With mock interviews and expert guidance

GROW
Be ready for opportunities in tech careers

SKILLS TODAY.
OPPORTUNITIES TOMORROW.

Gain the practical experience, portfolio and confidence to take the next step in your full-stack development career.



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TAKE THE NEXT STEP
IN YOUR FULLSTACK JOURNEY

Learn, build and gain real-world experience with Impara.
Join our 3-Month Full Stack Program and develop the skills to create smart, connected solutions.

PROGRAM DURATION
3 Months
(12 Weeks)

MODE
Online / Offline
(As per batch)

NEXT BATCH
September 2026

CERTIFICATION
Course completion certificate from Impara

READY TO GET STARTED?

Scan the QR code or fill out the form to register your interest. Our team will contact you with the detailed program information, batch schedule and next steps.

SCAN TO REGISTER

OR

Reply on WhatsApp for more details

Email us at
imaparacreativelabs@gmail.com

Call us
+91 98432 86522

LET'S BUILD A SMARTER TOMORROW.

Gain practical skills, build real-world projects and be a part of the next generation of innovators with Impara.



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Chennai, India
Working with students and institutions across India.

50+
Academic Collaborations

100+
Industry Projects

AICTE
NIRF Approved Internships